Prime63 m=2 diagnostic integration

TASK-20261018-7368da · 18 October 2026 · prospective engineering

Scientific runs in this handoff: 0. The proposed three-cell diagnostic remains unapproved. The archived m=2 positives were each censored at 300 seconds; this implementation prepares a bounded, phase-traced route for a future exact experiment. It carries no runtime or speedup claim.

Admission and custody

The parent and worker each check the pinned source manifest, committed frozen plan, repository authorization, owner and epoch, diagnostic entrypoints, and ready or exact in-flight selector before task output or Sage. The in-flight route binds the supervisor launch, trial, commands, run directory and launch age. A synthetic unreadable-live-root monitor control runs before all positives; failure prevents the positive calls.

Prospective limits: 600 seconds, 8 GiB sampled process-tree RSS, 8 GiB scratch and model cap 64 per call; 2100 seconds total, one worker, 16 GiB initial free scratch and stop below 8 GiB. First-attempt request, append-only phase/resource trace, stdout, stderr and response when present are hashed into the raw archive. Missing RSS is null. Time/resource caps are censored; monitor faults are infrastructure errors.

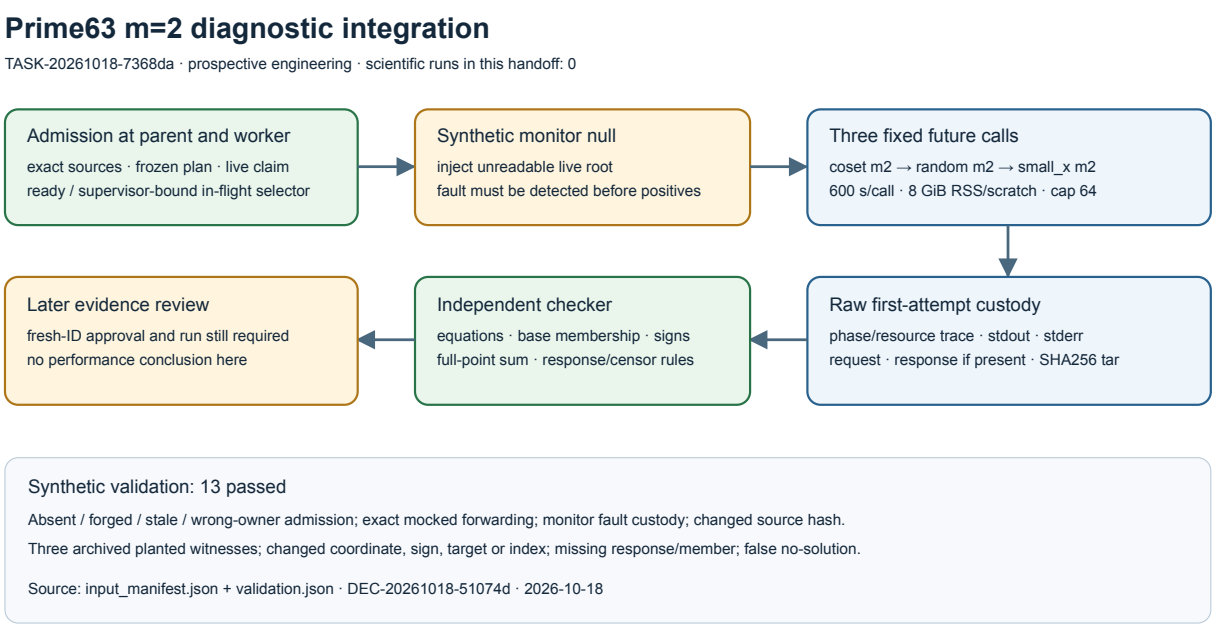
Independent checker

The checker evaluates every polynomial equation modulo the 63-bit prime; checks curve points, signed full-point sum, factor-base membership and summand positions; and verifies trace sequence, source hashes, raw archive coverage and response semantics. Archived witness indices denote summand positions, while the checker derives factor-base indices separately. A producer no-solution assertion contradicted by the independently supplied archived witness is rejected.

Synthetic validation

Thirteen tests passed. They cover absent and forged admission, stale supervisor launch, wrong owner, mocked exact worker forwarding, a synthetic monitor fault with retained trace and raw streams, source mismatch, three planted witnesses, changed coordinate/sign/target/index, missing response versus justified censor, missing raw member and false no-solution. The mocked worker makes no scientific solver call.

Flow and limits



The optional 475² ordered-pair full-point enumeration is a separately charged future control and was not run. A disk error before the monitor can create its output directory can leave no call archive; a future approved supervisor must retain that launch failure in its execution receipt. Sampled RSS does not bound an unsampled transient peak. A future scientific run requires fresh-ID preapproval, an approved committed plan, published claim and independent evidence review before any performance conclusion.

Evidence: input_manifest.json, diagnostic_packet.json, validation.json, execute.py, check.py, test_integration.py and DEC-20261018-51074d. Provenance: internal; exact static and synthetic checks by the Executor, 2026-10-18.